

Generalized linear models

Basic concepts of regression

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Basic concepts of regression - Contents

Components of regression

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Components of regression

Fundamental problem

How do variables $x_1, \dots, x_p$ affect a dependent variable y ?

$x_1, \dots, x_p$	$\rightarrow$	y
covariates		outcome
influence variables		target (variable)
explanatory — " —		response — " —
independent — " —		dependent — " —

Components of regression

Questions:

1. What is the difference to correlation?

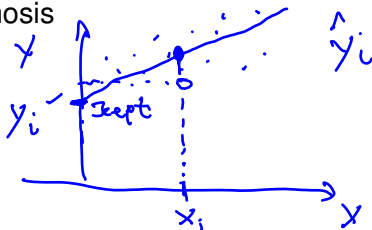
- direction of the investigated relationship
- multiple influence variables

2. What is the difference to statistical testing?

- influence of several quantities/variables at the same time / simultaneously
- predictions on new observations (e.g. $\hat{y}$)

What are the main objectives?

- Detection of interdependency
clinical outcome: Blood pressure
therapy: new drug
Severity of the disease
- Quantification of the relation
clinical outcome
dose of the drug
Baseline: Severity
- Prediction / prognosis



What is being investigated?

$$y | \underline{x} = \underline{x}$$

The variable $y | \mathbf{x}$ with $\mathbf{x} = (x_1, \dots, x_p)^\top$ is examined.

Particularly:

► $f_{y|\mathbf{x}}$ or $f_y(\mathbf{x})$

cond. density

► $F_{y|\mathbf{x}}$ or $F_y(\mathbf{x})$

cond. distribution function

► $\mathbb{E}_{y|\mathbf{x}}$ or $\mathbb{E}(y | \mathbf{x})$

cond. expected value

► $Q_T(y | \mathbf{x})$

cond. quantiles

Structures

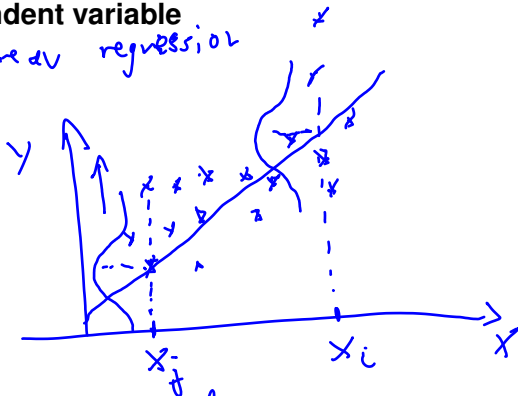
Structure of the dependent variable

- y metrical

linear regression

often assumption

$$y_i | x_i \sim \mathcal{N}(\mu_i, \sigma^2)$$

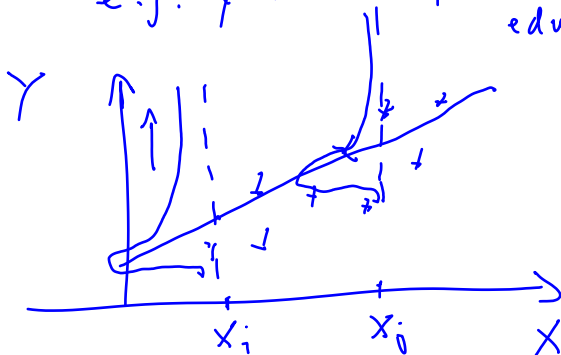


e.g. influence of drug dose on
systolic blood pressure

Structures

- Special case: y skewed distribution

e.g. y income, x number of years in education system

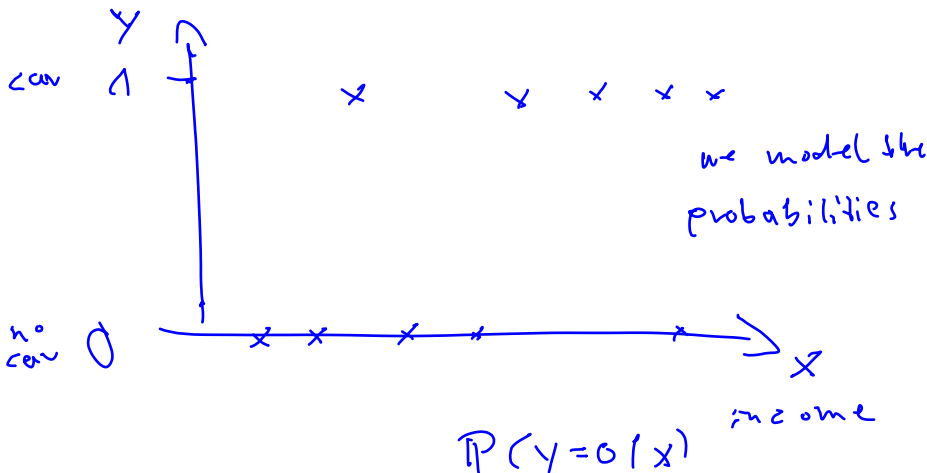


$$E[Y | x_i]$$

Strukturen

$$P(y=1|x)$$

► y binary



Strukturen

- y multi-categorical

e.g. diabetes as a consequence of overweight

y : - no diabetes (1)

- Type I diabetes (2)

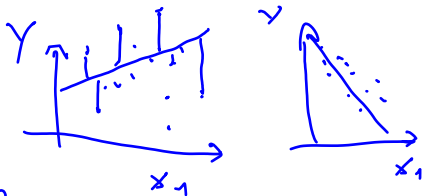
- Type II diabetes (3)

- in addition, depend on risk factors, such as age, gender, BMI, ...

Structures

Structure of the explanatory term

- Simplest case: linear



$$E[Y | X] = \beta_0 + X^T \beta$$

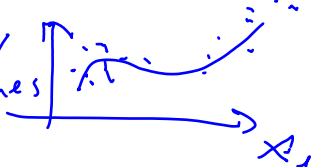
$$= \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$$

- More general: additive models

$$E[Y | X] = \beta_0 + f_1(x_1) + \dots + f_p(x_p):$$

$f_i(\cdot)$: - have arbitrary form
+ parametric (e.g. x^2), splines

But: always additive



Structures

main effects

- Additive model with interactions



$$E[y | \underline{x}] = \beta_0 + f_1(x_1) + f_2(x_2) + \underbrace{f_{1,2}(x_1, x_2)}$$

interaction term

- Additive model with an effect-modified variable

$$E[y | \underline{x}] = \beta_0 + f_2(x_2) + x_1 f_{1,2}(x_2)$$

E.g. x_1 : gender

x_2 : age

effect of age is influenced/modified by gender

Encoding of covariates

$$\beta_1 x_1 \rightarrow \beta_1 (x_1 + 1) = \beta_1 x_1 + \underline{\beta_1}$$

Linear model

$$\mathbb{E}(y \mid \mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

1. x_j metrical, e.g.:

► y : reaction time

► x_1 : age

► β_1 : 20 $\frac{\text{msec}}{\text{year}}$

Interpretation: reaction time is increased by
20 msec per year of age

$x_j \beta_j$

For multiple independent variables

- ▶ One independent variable, e.g.
 - ▶ y : net rent
 - ▶ x_1 : number of rooms
 - ▶ β_1 : 188,58 € / room
- ▶ Two independent variables, e.g.
 - ▶ y : net rent
 - ▶ x_1 : # rooms
 - ▶ x_2 : size of apartment
 - ▶ β_1 : -23,12 € / room
 - ▶ β_2 : 7,841 € / m²

For multiple independent variables

- β_j is interpretable given other variables are fixed (adjustment)

$$\begin{aligned} & E[y | x_1, \dots, x_j + 1, \dots, x_p] - E[y | x_1, \dots, x_j, \dots, x_p] \\ &= \beta_0 + \dots + (x_j + 1)\beta_j + \dots + x_p\beta_p - \beta_0 - \dots - x_j\beta_j - \dots - x_p\beta_p \\ &= (x_j + 1)\beta_j - x_j\beta_j = \beta_j \end{aligned}$$

- marginal correlation $\neq$ conditional correlation

→ coefficients change, if ^{new} variables are included

When is this not the case? If a new variable has exactly no effect (theoretically)

Encoding of covariates

$\beta_0 \hat{=}$ average reaction time of females

2. x_j binary

- ▶ y : reaction time
- ▶ x_1 : gender $\begin{cases} 1 \text{ male} \\ 0 \text{ female} \end{cases}$
- ▶ $\beta_1 = 30,3 \text{ msec}$

Interpretation:
On average males have an ^{increased} reaction time of 30,3 msec more than females.

Encoding of covariates

3. x_j multiple-categorical, e.g.

► y : reaction time

► x_1 : time of day

1	morning
2	noon
3	evening

or, alternatively

1	male
2	female
3	children

$\Rightarrow E[y | x] = \beta_0 + \beta_1 x_1$ not reasonable,
as distances are not comparable, and
interchangeable

$\Rightarrow$ solution: encoding

Dummy coding: 0/1-Encoding

- ▶ $x_j \in 1, \dots, k$
- ▶ $x_{j(i)} = \begin{cases} 1 & , x_j = i \\ 0 & , \text{otherwise} \end{cases} \quad , i = 1, \dots, k$

x_j	1	$x_{j(1)}$	$x_{j(2)}$	...	$x_{j(k)}$
1	1	1	0	...	0
2	1	0	1	...	0
$\vdots$	1	$\vdots$	$\vdots$		$\vdots$
k	1	0	0		1
	1				

Dummy coding: 0/1-Encoding

► Model: $\mathbb{E}(y | x_j) = \beta_0 + \beta_{j(1)}x_{j(1)} + \dots + \beta_{j(k)}x_{j(k)}$

► Problem:

$k+1$ parameters $\beta_0, \beta_{j(1)}, \dots, \beta_{j(k)}$

for k expected values,

$$\mathbb{E}[y | x_j = 1], \dots, \mathbb{E}[y | x_j = k]$$

$\Rightarrow \beta$'s are not unique anymore

$\Rightarrow$ model is "not identifiable"

$\rightarrow$ (design matrix does not have full rank $\Rightarrow$ not invertible)

Dummy coding: Identifiability

Reference coding

- ▶ To achieve uniqueness (identifiability) of the parameters, restrictions are required
- ▶ E.g., one category is set to 0
- ▶ Convention:
 - ▶ Set category k to 0 $\rightarrow \beta_{j(k)} = 0$
 $\rightarrow$ reference category

$$\rightarrow \beta_{j(i)} = 0$$

or choose the largest category

$$E[y | x_j] = \beta_0 + \beta_{j(1)} x_{j(1)} + \dots + \beta_{j(h-1)} x_{j(h-1)}$$

$\Rightarrow x_{j(k)}$ has been removed, as we chose k to be the reference cat.

Dummy coding: Identifiability

Example: $k = 3$

$$\begin{cases} 1, & x_1 = 2 \\ 0, & \text{otherwise} \end{cases}$$

$$\mathbb{E}(y | x_1) = \beta_0 + \beta_{1(1)}x_{1(1)} + \beta_{1(2)}x_{1(2)} + \beta_{1(3)}x_{1(3)}$$

$$= 0$$

► $\mathbb{E}(y | x_1 = 1) = \beta_0 + \beta_{1(1)}$

► $\mathbb{E}(y | x_1 = 2) = \beta_0 + \beta_{1(2)}$

► $\mathbb{E}(y | x_1 = 3) = \beta_0$

⇒ β_0 is expected value of the reference cat.

$$\Rightarrow \beta_{1(i)} = \mathbb{E}[y | x_1 = i] - \mathbb{E}[y | x_1 = \text{ref}]$$

⇒ Diff. to the reference

Effect encoding

- ▶ $x_j \in 1, \dots, k$
- ▶ Model: $\mathbb{E}(y \mid x) = \beta_0 + \beta_{j(1)}x_{j(1)} + \dots + \beta_{j(k)}x_{j(k)}$
- ▶ **Problem:** again too many parameters !

Effect encoding: Identifiability

introduce side constraint

- ▶ Restriction: $\sum_{i=1}^k \beta_{j(i)} = 0 \Leftrightarrow \beta_{j(k)} = -\beta_{j(1)} - \dots - \beta_{j(k-1)}$
- ▶ We have:

$$\frac{1}{k} \sum_{i=1}^k E[y | x_j = i]$$

$$= \frac{1}{k} \sum_{i=1}^k (\beta_0 + \beta_{j(i)}) = \frac{1}{k} k \cdot \beta_0 + \frac{1}{k} \underbrace{\sum_{i=1}^k \beta_{j(i)}}_{=0} = \beta_0$$

→ β_0 is representing the mean over all categories, i.e. the average effect

Effect encoding

Representation:

$$x_{j(i)} = \begin{cases} 1 & , \quad x_j = i \\ -1 & , \quad x_j = k \\ 0 & , \quad \text{otherwise} \end{cases} \quad i \in \{1, \dots, k-1\}$$

$$\mathbb{E}(y \mid x) = \beta_0 + \beta_{1(1)}x_{1(1)} + \dots + \beta_{1(k-1)}x_{1(k-1)}$$

Then:

$$\triangleright \mathbb{E}(y \mid x_1 = 1) = \beta_0 + \beta_{1(1)}$$

$$\vdots$$
$$\triangleright \mathbb{E}(y \mid x_1 = k-1) = \beta_0 + \beta_{1(k-1)}$$

$$\triangleright \mathbb{E}(y \mid x_1 = k) = \underbrace{\beta_0 - \beta_{1(1)} - \dots - \beta_{1(k-1)}}_{= \beta_{j(k)}}$$

only up to $k-1$

Dummy and effect encoding

Which encoding is useful in which situation?

- ▶ Dummy coding

If the difference to some specific reference is relevant, e.g. when there is a standard therapy or placebo/no therapy

- ▶ Effect encoding

If changes from the overall mean over all categories are of interest

⇒ makes no difference for the modelled means, i.e. $E[y | x=i], i=1, \dots, k$

Linear predictor

→ in the linear model

$$\eta = \mathbb{E}[\gamma | \mathbf{x}]$$

→
$$\eta = \mathbf{x}^T \boldsymbol{\beta} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

Variables can

- ▶ be metrical, encoded or both
- ▶ include polynomial terms etc., e.g. age^2

Predictor characterization via operators/ Wilkinson-Rogers notation

Let $x, \tilde{x}$ be metrical, and j be categorical

Operator	Abbreviation	Linear term
metric / binary	$\rightarrow x$	$x\beta$
factors	$\rightarrow j$	$x_{j(1)}\beta_{j(1)} + \dots + x_{j(k-1)}\beta_{j(k-1)}$
+	$x + \tilde{x}$	$x\beta_x + \tilde{x}\beta_{\tilde{x}}$
.	$x + j$	$x\beta_x + x_{j(1)}\beta_{j(1)} + \dots$
.	$x \cdot \tilde{x}$	$x \cdot \tilde{x} \beta_{x\tilde{x}}$
*	$x \cdot j$	$x x_{j(1)}\beta_{xj(1)} + x x_{j(2)}\beta_{xj(2)} + \dots$
	$x * \tilde{x}$	" $x + \tilde{x} + x \cdot \tilde{x}$ "
	$x * j$	" $x + j + x \cdot j$ "

Classical linear regression model

Data $(y_1, x_1), \dots, (y_n, x_n)$ unknown

Linear regression model

$$y_i = \mathbf{x}_i^\top \beta + \varepsilon_i \quad i=1, \dots, n$$

RV determ. fixed RV

with

1. $\mathbb{E}(\varepsilon_i) = 0$
 2. $\text{Var}(\varepsilon_i) = \sigma^2$
 3. $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$
 4. $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$
- $\Rightarrow \mathbb{E}[y_i | x_i] = x_i^\top \beta$
 $\Rightarrow \text{Var}(y_i | x_i) = \sigma^2$
 $\Rightarrow$ errors independent
 $\Rightarrow$ for testing

Classical linear regression model - alternative representation

- Random component:

the errors ε_i
have
disappeared

mean dep. on i
variance homogeneous

$$y_i | \mathbf{x}_i \sim \mathcal{N}(\mu_i, \sigma^2)$$

distribution

- Systematical component:

How does $\mathbf{x}_i$ determine the mean

$$\mu_i = \mathbb{E}(y_i | \mathbf{x}_i) = \mathbf{x}_i^\top \boldsymbol{\beta} \Rightarrow \eta_i$$

→ our model

→ we will model the mean